

Named Entity Recognition

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1 Introduction

Named entity recognition (NER) aims at automatically detecting and categorizing named entities within spans of words. While this concept is quiet easy to grasp, solving this problem automatically has become an important area of research as annotation by hand is fairly time-consuming. Researchers have tackled this challenge with a variety of AI-model-architectures reaching from simple RNNs over LSTMs to the current state-of-the-art: Transformers. Finding methods to further improve the results has lead to the development of many different approaches, among them data preprocessing, feature engineering and changes to the AI-models themselves. In this project some similar improvement attempts were implemented, motivated by the literature shortly described in section 2.

2 Related work

Among the transformer models released over the course of years the performance of Bidirectional Encoder Representations from Transformers (BERT) in NLP-related tasks significantly stands out (Devlin et al., 2019). BERT is a language representation model that performs pre-training on bidirectionally represented, unlabeled text data. In the next step, it can be fine-tuned on a variation of datasets for a multitude of tasks. As a result, the base architecture is allowed to remain the same and does not have to be task-specific (Devlin et al., 2019). Therefore, leveraging its performance for NER is an obvious choice that was made by many researchers. Some of their approaches aim to build a better performing model based on BERT.

Souza, Nogueira, and Lotufo train a Portuguese BERT-model and combine it with a conditional random field (CRF) for named entity recognition. The tokenized words are used as input for BERT. Thereafter, its output is classified by a linear layer, whose output consisting of sequences of tag scores is then fed into the CRF-layer in order to obtain the final labels (Souza, Nogueira, and Lotufo, 2020). Similarly to Souza, Nogueira, and Lotufo, Li et al. make changes to their model for NER. They add a bidirectional Long-Short-Term-Memory-layer (BiLSTM) in between the BERT-model and a CRF-layer, which predicts the probability distributions of entity labels (Li et al., 2022).

While some approaches make changes to the model, others include features in their training and fine-tuning procedure of BERT. Part-of-speech-tags (POS-tags) are among the more frequently used features since they capture syntactical meaning of words. 'walk', for example, can represent the verb 'to walk' or the noun 'a walk'. Another commonly used feature are syntactic dependencies, which capture relationships between words. Speech is ambiguous by nature, so multiple existing approaches include the formerly mentioned features to mitigate this. While Syed and Chung incorporate POS-tags as features for training their BERT-model for NER, Zhang et al. additionally make use of other linguistic features such as dependencies. While the latter approach tackles a question answering task, more precisely the SQuAD 2.0 challenge, which has no direct connection to NER, their use of dependencies might be applicable to NER. In order to make use of POS-tags and dependencies, they combine their BERT-backbone-model with a linear layer that incorporates the formerly mentioned linguistic features. As these features consist of strings, each of them is assigned to ids in the form of integer values (Zhang, 2019).

Preprocessing the input data is another way to improve the results of NER. Keretna, Lim, and Creighton, for example, remove stop words such as 'and' or 'has' from their dataset since they carry little to no meaning. Munkhdalai et al., too, remove 'non-informative characters' (Munkhdalai et al., 2012) before performing NER on biomedical data.

All of these approaches have a common goal: Improving named entity recognition. Motivated by their results, this project implements a combination of several methods described in detail in the following section.

3 Method

3.1 Data Preprocessing

The dataset consists of a training dataset containing 68 abstracts of medical data and a test dataset with 20 similarly sourced abstracts. All sentences of the training and test dataset are loaded as lists. The corresponding labels which indicate the position and class of named entities are inserted at their respective indices. The remaining, unlabeled word's labels are denoted as 'O'. In a second step, the formerly made labels are extended to match the 'BIO'-tagging scheme. The first label belonging to a named entity is extended with a 'B-' at its start, while its following labels belonging to the same entity are extended with an 'I-'. Every word that has no label gets labeled with 'O'. The part of 1 that was marked in blue shows an example sentence with NER-labels. In addition to the original

Sentence	One	fracture	occurred	in	each	treatment	group	.
NER-labels	B-NumAffected	O	O	O	O	O	O	O
POS-tags	NUM	NOUN	VERB	ADP	DET	NOUN	NOUN	PUNCT
Dependencies	nummod	nsubj	ROOT	prep	det	compound	probj	punct

Figure 1: Example sentence from the original dataset with NER-tags in BIO-labelling scheme (blue) with additional POS-tags and syntactical dependencies (green).

dataset, a filtered version was made where words consisting of a single characters and words containing punctuation were deleted from the dataset. This will henceforth be referred to as 'filtered dataset'.

The original dataset solely provides a training and a test dataset. However, training neural networks often benefits from an additional validation dataset. Therefore, about 20% of the training dataset are cut off and used as validation dataset for training the models described in section 3.3.

As a last step before training, the sentences are padded to the length of the longest sentence and tokenized using word ids for the BiLSTM-model and the BERT-Tokenizer for the BERT-models and its variants. The latter one automatically produces sub-word-tokens. The tokenized words of the BiLSTM-model are then transformed to vector embeddings using gensim (Rehurek and Sojka, 2011).

3.2 Feature Engineering

In order to make use of syntactical information during training, two additional features are leveraged: POS-tags and syntactic dependencies. Both feature vectors are automatically generated using the spacy library (Honnibal and Montani, 2017). An example sentence with both features is shown in the part of Figure 1 that is marked in green.

3.3 Models

NER was performed using a bidirectional LSTM as simple neural baseline, a fine-tuned BERT-model and variations of the BERT-model.

1. Simple neural baseline: Bidirectional LSTM with learning rate 0,01 over 50 epochs with early stopping enabled.
2. LLM-finetuning-approach: Devlin et al. tested BERT over several tasks and found some parameters that tended to work well in most cases. They suggest to use 5e-5, 3e-5 or 2e-5 as learning rate with the Adam optimizer and fine-tune the model for 2, 3, or 4 epochs on batch sizes 16 or 32. Thus, this model (bert-base-cased) was trained with a learning rate of 2e-5 over 3 epochs.
3. Extensions of BERT: BERT-CRF and BERT-BiLSTM-CRF As suggested by multiple approaches - some of them were described in Section 2 - a CRF layer (a) and a bidirectional LSTM layer followed by a CRF layer (b) were added to the BERT-model (bert-base-cased). In order to combine BERT with additional input features, a third model with two feature embeddings (c) was deployed as well. A short, written overview of the model layers and dimensions with batch-size 1 is given below followed by a graphical representation with batch size 16.

(a) BERT-CRF: $Input(128) \rightarrow BERT(128, 768) \rightarrow Dropout(128, 768) \rightarrow Linear(128, 65) \rightarrow CRF(128, 1)$

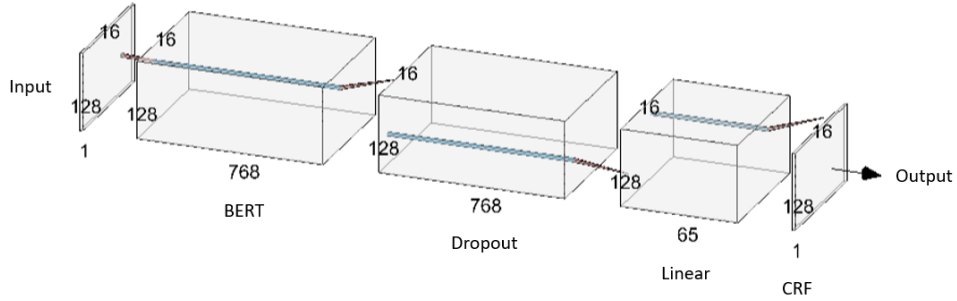


Figure 2: Example visualization of the BERT-CRF-model.

(b) BERT-BiLSTM-CRF: $Input(128) \rightarrow BERT(128, 768) \rightarrow Dropout(128, 768) \rightarrow BiLSTM(128, 768) \rightarrow Linear(128, 65) \rightarrow CRF(128, 1)$

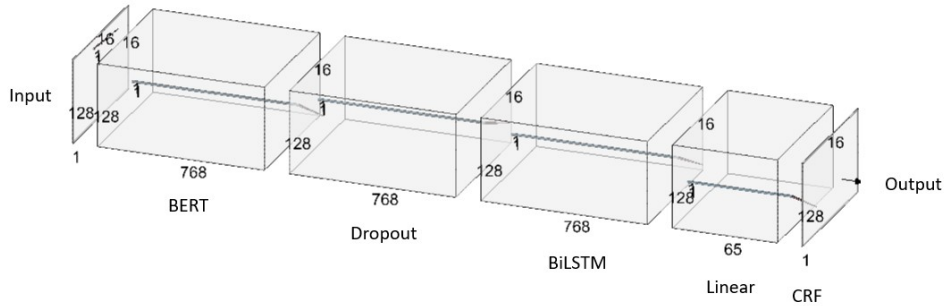


Figure 3: Example visualization of the BERT-BiLSTM-CRF-model.

(c) BERT-CRF with features: $Input(128) \rightarrow BERT(128, 768) \rightarrow Embedding(128, 768) \rightarrow Embedding(128, 768) \rightarrow Dropout(128, 768) \rightarrow Linear(128, 65) \rightarrow CRF(128, 1)$

Training was performed with learning-rate 4e-5 over 16 epochs which is in line with the approaches mentioned in section 2

The BERT variations described in (2) and (3) were subsequently trained and tested in multiple combinations with the original dataset, the filtered dataset and additional feature embeddings. The results can be found in the following section.

4 Results

The BiLSTM-model that is used as simple neural baseline yields an over 5 training runs averaged Micro-F1-Score on the test set of 82,63% which is worse than the results of the BERT-model. Thus, further experiments were conducted with the BERT-model. The following table shows the results of all experiments conducted with different combinations of data filtering, the addition of input features and variations of the BERT-model. The results are, again, the Micro-F1-scores of the test set averaged over 5 training runs. Since the results of the BERT-BiLSTM-CRF-model are significantly worse and

	BERT	BERT-CRF	BERT-CRF-BiLSTM
data unchanged, no features	84,04%	83,93%	70,46%
filtered data	86,06%	84,43%	73,38%
with features	-	84,72%	-
filtered data, with features	-	84,2%	-

tend to fluctuate a lot (approx. 10%), these values are reported, but remain greyed out. Consequently, no further experiments beyond data filtering were conducted with this model type. A visualization of these results can be found in the figure below.

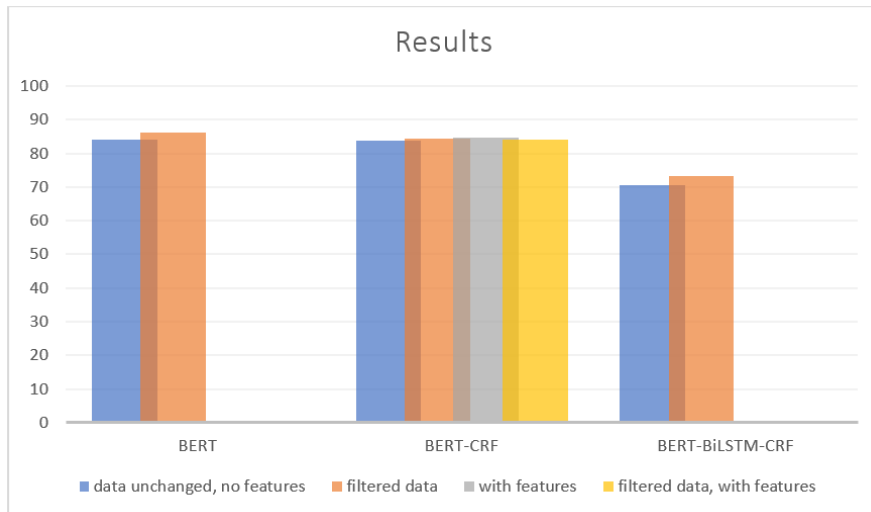


Figure 4: Bar chart of experiment results.

The scores of BERT and BERT-CRF yield very similar values, with BERT-CRF combined with features showing the best results out of this model type, beating the unmodified BERT baseline by 0,7%. However, not only is the training of the BERT-model alone much faster as it only runs for 4 epochs instead of 16, its variation with filtered data beats out all results of BERT-CRF as well.

5 Conclusion

In this project, NER is performed on medical data using a bidirectional LSTM, a fine tuned BERT-model and a BERT-model that was altered by incorporating a CRF-layer and another variation with an additional BiLSTM- and CRF-layer. Fine-tuning BERT alone results in a Micro-F1-score of 84,04% on the test set. Outperforming this state-of-the-art model has proven itself to be a challenging task and only succeeded by a fairly small margin. In the future, other pretrained transformer models could be trained for the same task on the same dataset to find an even better one. Furthermore, other feature engineering techniques, such as lemmatization and stemming, could be applied.

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